

1 Solution — GIAM 5.3

Problem 7

1.1 Problem

Give an inductive proof that $\forall n \in \mathbb{N}, 4 \mid 13^n + 4n - 1$.

1.2 Answer

$$f(k+1) = f(k) + 4(3 \cdot 13^k + 1)$$

Here $f(n) = 13^n + 4n - 1$. The step adds a multiple of 4, so divisibility passes from k to $k+1$. Verified for every n up to 3000.

1.3 The Proof

Theorem. For every natural number n , 4 divides $f(n) = 13^n + 4n - 1$.

Proof. By induction on n .

Basis. At $n = 0$, $f(0) = 1 + 0 - 1 = 0$, and $4 \mid 0$.

Inductive step. Let k be a particular but arbitrarily chosen natural number and suppose $f(k) = 4m$ for some integer m . Then

$$\begin{aligned} f(k+1) &= 13 \cdot 13^k + 4(k+1) - 1 \\ &= (13^k + 4k - 1) + 12 \cdot 13^k + 4 \\ &= f(k) + 4(3 \cdot 13^k + 1). \end{aligned}$$

Substituting the hypothesis,

$$f(k+1) = 4(m + 3 \cdot 13^k + 1),$$

and the bracket is an integer, so $4 \mid f(k+1)$.

Basis and step together give the result for every n . ■

The key move is splitting $13 \cdot 13^k$ as $13^k + 12 \cdot 13^k$, which puts a copy of $f(k)$ inside $f(k+1)$. Another route multiplies the hypothesis by 13 instead, giving $f(k+1) = 13f(k) - 16(3k-1)$; that works too, and its 16 is a first hint of the second remark below.

1.4 The Picture

The values turn out to be multiples of 16, and the residues modulo 16 show why.

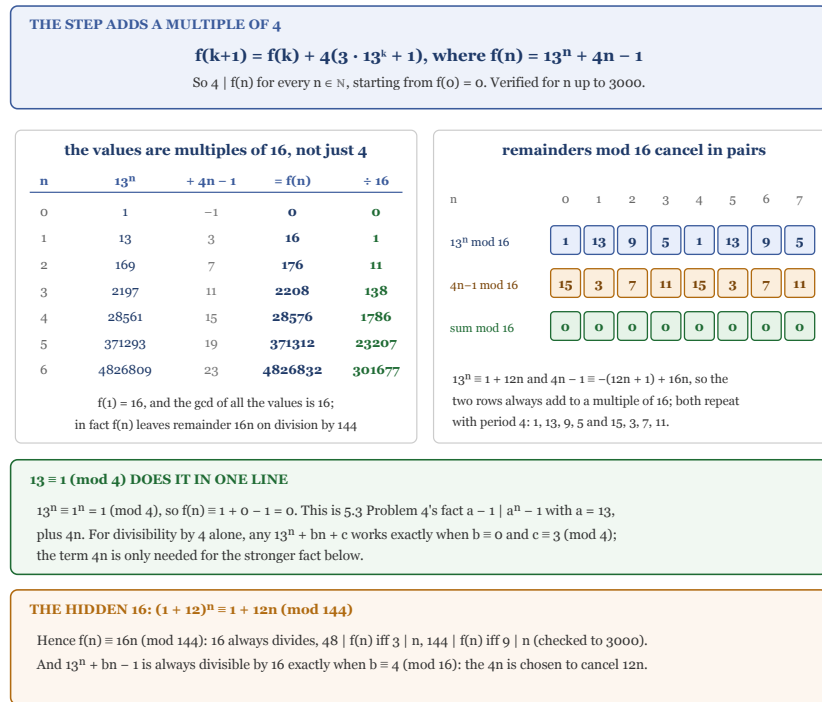


Figure 1.1: A diagram in four parts. The top band states the step identity f of k plus one equals f of k plus four times the quantity three times thirteen to the k plus one, where f of n is thirteen to the n plus four n minus one. The left panel tabulates, for n from zero to six, thirteen to the n , four n minus one, and f of n , namely 0, 16, 176, 2208, 28576, 371312, 4826832, together with f of n over sixteen: 0, 1, 11, 138, 1786, 23207, 301677. The right panel lists, for n from zero to seven, thirteen to the n mod sixteen (1, 13, 9, 5 repeating) and four n minus one mod sixteen (15, 3, 7, 11 repeating), whose sums are all zero mod sixteen. A green band gives the one-line proof from thirteen congruent to one mod four. An orange band states that thirteen to

the n is congruent to one plus twelve $n \bmod 144$, so f of n is congruent to sixteen $n \bmod 144$, and that the coefficient four is exactly what makes sixteen divide.

1.5 Remark — The One-Line Congruence Proof

Since $13 \equiv 1 \pmod{4}$, every power of 13 is also $\equiv 1$, and so

$$f(n) = 13^n + 4n - 1 \equiv 1 + 0 - 1 = 0 \pmod{4}.$$

This is 5.3 Problem 4's fact $a - 1 \mid a^n - 1$ with $a = 13$, which gives $12 \mid 13^n - 1$ and therefore $4 \mid 13^n - 1$, with the harmless multiple $4n$ added on. For divisibility by 4 alone the linear term is irrelevant: a search over residues finds that $13^n + bn + c$ is always divisible by 4 exactly when $b \equiv 0$ and $c \equiv 3 \pmod{4}$. So why $4n$? The next remark answers that.

1.6 Remark — The Hidden 16

The table shows every value is a multiple of 16 , and the gcd of $f(1), \dots, f(299)$ is exactly 16 (it cannot be more, since $f(1) = 16$). The binomial theorem explains it. Writing $13 = 1 + 12$,

$$13^n = (1 + 12)^n = 1 + 12n + \binom{n}{2}12^2 + \binom{n}{3}12^3 + \dots,$$

and every term after the second is a multiple of $144 = 12^2$. So $13^n \equiv 1 + 12n \pmod{144}$, checked to $n = 3000$, and hence

$$f(n) \equiv 1 + 12n + 4n - 1 = 16n \pmod{144}.$$

This one congruence gives a lot. Since $16 \mid 144$, it shows $16 \mid f(n)$ for every n . It also shows $48 \mid f(n)$ exactly when $3 \mid n$ and $144 \mid f(n)$ exactly when $9 \mid n$, both checked to $n = 3000$.

Now the role of $4n$ is clear. Modulo 16 , $13^n - 1 \equiv 12n$, so

$$16 \mid 13^n + bn - 1 \text{ for every } n \iff b \equiv 4 \pmod{16},$$

and a search over b from 0 to 39 finds exactly 4, 20, 36. The $4n$ is precisely the term that cancels $12n$ modulo 16. The problem asks for 4, but it was built for 16.

1.7 Remark — The General Pattern Behind the $4n$

The congruence used above is one case of

$$(1 + d)^n \equiv 1 + dn \pmod{d^2},$$

which holds for every integer d by the same binomial expansion; it was checked for d from 2 to 30 and n up to 200. It is the discrete shadow of the approximation $(1 + x)^n \approx 1 + nx$ for small x , with d^2 playing the role of the negligible x^2 terms.

Applied to 5.3 Problem 4 with $4 = 1 + 3$, it gives $4^n \equiv 1 + 3n \pmod{9}$, so $9 \mid 4^n - 3n - 1$ for every n (checked to $n = 500$). That is a companion exercise of exactly this problem's shape: a power, a linear correction, and a constant, with the correction chosen to cancel the dn term.